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Robotic surgical system with virtual control panel for tool actuation

Abstract

A surgical system includes a detector, comprising an array of pixels configured to detect light reflected by a surgical instrument and generate a first signal comprising a first dataset representative of a visible image of the surgical instrument. The surgical system also includes a processor configured to receive the first signal, generate a modified image of the surgical instrument that includes a control panel. The control panel includes one or more control elements representative of one or more operating parameters of the surgical instrument. The processor is further configured to receive an input to the control panel from a user, the input being effective to change one of the operating parameters. The processor is also configured to generate a command signal based on the input to change the one of the operating parameters.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional of U.S. Ser. No. 16/783,656 (now U.S. Pat. No. 11,547,494) filed Feb. 6, 2020, entitled “ROBOTIC SURGICAL SYSTEM WITH VIRTUAL CONTROL PANEL FOR TOOL ACTUATION,” which is a divisional of U.S. patent application Ser. No. 15/383,026 (now U.S. Pat. No. 10,568,701) filed Dec. 19, 2016, entitled “ROBOTIC SURGICAL SYSTEM WITH VIRTUAL CONTROL PANEL FOR TOOL ACTUATION,” which are hereby incorporated by reference in their entireties.

FIELD

(1) Methods and devices are provided for minimally invasive surgery, and in particular for providing virtual control panel of a surgical instrument.

BACKGROUND

(2) Minimally invasive surgical (MIS) instruments are often preferred over traditional open surgical instruments due to the reduced post-operative recovery time and minimal scarring. Laparoscopic surgery is one type of MIS procedure in which one or more small incisions are formed in the abdomen and a trocar is inserted through the incision to form a pathway that provides access to the abdominal cavity. The trocar is used to introduce various instruments and tools into the abdominal cavity, as well as to provide insufflation to elevate the abdominal wall above the organs. The instruments and tools can be used to engage and/or treat tissue in a number of ways to achieve a diagnostic or therapeutic effect. Endoscopic surgery is another type of MIS procedure in which elongate flexible shafts are introduced into the body through a natural orifice.

(3) Although traditional minimally invasive surgical instruments and techniques have proven highly effective, newer systems may provide even further advantages. For example, traditional minimally invasive surgical instruments often deny the surgeon the flexibility of tool placement found in open surgery. Difficulty is experienced in approaching the surgical environment with the

instruments through the small incisions. Additionally, the added length of typical endoscopic instruments often reduces the surgeon's ability to feel forces exerted by tissues and organs on the end effector. Furthermore, coordination of the movement of the end effector of the instrument as viewed in the image on the television monitor with actual end effector movement is particularly difficult, since the movement as perceived in the image normally does not correspond intuitively with the actual end effector movement. Accordingly, lack of intuitive response to surgical instrument movement input is often experienced. Such a lack of intuitiveness, dexterity, and sensitivity of endoscopic tools has been found to be an impediment in the increased use of minimally invasive surgery.

(4) Over the years a variety of minimally invasive robotic systems have been developed to increase surgical dexterity as well as to permit a surgeon to operate on a patient in an intuitive manner. Telesurgery is a general term for surgical operations using systems where the surgeon uses some form of remote control, e.g., a servomechanism, or the like, to manipulate surgical instrument movements, rather than directly holding and moving the tools by hand. In such a telesurgery system, the surgeon is typically provided with an image of the surgical environment on a visual display at a location remote from the patient. The surgeon can typically perform the surgical procedure at the location remote from the patient whilst viewing the end effector movement on the visual display during the surgical procedure. While viewing typically a three-dimensional image of the surgical environment on the visual display, the surgeon performs the surgical procedures on the patient by manipulating master control devices at the remote location, which master control devices control motion of the remotely controlled instruments.

(5) While significant advances have been made in the field of minimally invasive surgery, there remains a need for improved methods, systems, and devices for providing virtual control panel of a surgical instrument.

SUMMARY

(6) Methods, devices, and systems are provided for displaying a modified image of a surgical environment wherein the modified image includes at least a portion of a surgical instrument and a control panel associated with the surgical instrument. The control panel displayed in the modified image is active in that functions associated with the surgical instrument can be actuated by interacting with various specific elements of the control panel displayed in the modified image.

(7) A surgical system includes a detector comprising an array of pixels configured to detect light reflected by a surgical instrument and generate a first signal comprising a first dataset representative of a visible image of the surgical instrument. The surgical system also includes a processor configured to receive the first signal, generate a modified image of the surgical instrument that includes a control panel that includes one or more control elements representative of one or more operating parameters of at least one of the surgical system and the surgical instrument. The processor is further configured to receive an input to the control panel from a user, the input being effective to change one of the operating parameters. The processor is also configured to generate a command signal based on the input to the control panel to change one of the operating parameters.

(8) The surgical system can vary in a number of ways. In one embodiment, the modified image is displayed on a display device. In another embodiment, the processor identifies from the first dataset data representative of the surgical instrument based on position of the detector with respect to the surgical instrument.

(9) In one embodiment, at least one of the position and shape of the control panel in the modified image is adjusted based on the orientation of the surgical instrument. In another embodiment at least one of the position and shape of the control panel with respect to the surgical instrument remains fixed.

(10) The surgical system includes an input device to receive the input from the user, and in one embodiment the input device is a user interface. In one embodiment, the processor generates a

pointer that moves relative to the modified image based on the input by the user. In one embodiment the command signal is generated when the user actuates one of the one or more control elements using the user interface device.

(11) The processor is also configured to receive a second signal representative of information related to the surgical instrument. The control panel is generated based on the second signal and data related to the surgical instrument that is stored in a database.

(12) The surgical instrument can be any type of instrument that is used in a surgical procedure. In one embodiment the surgical instrument is one that includes functionalities including at least one of tissue cutting, tissue stapling, and delivery of energy to tissue.

(13) In another aspect, a robotic surgical system includes a robotic arm, and a tool assembly removably coupled to the robotic arm and including a shaft extending distally from a housing and an end effector coupled to a distal end of the shaft, the end effector being configured to treat tissue. The robotic surgical system also includes a camera system configured to capture light reflected by at least one of the end effector and the shaft and generate a signal representative of a visible image of the at least one of the end effector and shaft. The robotic surgical system further includes a processor configured to receive the signal and generate a modified image of the at least one of the end effector and shaft that includes a control panel. The control panel in the modified image comprises one or more control elements representative of one or more operating parameters of the robotic surgical system. The processor is configured to receive an input to the control panel from a user, the input being effective to change one of the operating parameters, and generate a command signal based on the input to change the one of the operating parameters.

(14) In one embodiment, the robotic surgical system includes a display device, which can be wall or table mounted or on an accessory (e.g., head set, glasses, etc.), configured to display a modified image of the end effector and the shaft. The robotic surgery system also includes an input device to receive the input from the user. In one embodiment the input device is a user input device. In another embodiment, the processor generates a pointer that moves relative to the modified image based on the input by the user.

(15) In another aspect a surgical method includes receiving a first signal representative of an image of a surgical instrument and generating a modified image of the surgical instrument that includes a control panel. The control panel comprises one or more control elements representative of one or more operating parameters of at least one of the surgical system and the surgical instrument. The surgical method also includes receiving an input to the control panel from a user, the input being effective to change one of the operating parameters. The surgical method further includes generating a command signal based on the input to change the one of the operating parameters.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates a perspective view of an embodiment of a surgical robotic system that includes a patient-side portion and a user-side portion;

(3) FIG. 2 illustrates a schematic view of a virtual control panel generation system;

(4) FIG. 3 illustrates a schematic view of a detector in a camera module of the virtual control panel generation system;

(5) FIG. 4 illustrates an example of a virtual control panel of the system described herein;

(6) FIG. 5 illustrates an example of a modified image of a surgical instrument that includes the virtual control panel;

(7) FIG. 6 illustrates a surgical instrument positioned in a field having multiple markers; and

(8) FIG. 7 is a schematic illustration of a computer system configured to generate a plurality of command signals for use with the control system described herein.

DETAILED DESCRIPTION

(9) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

(10) Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape. Sizes and shapes of the systems and devices, and the components thereof, can depend at least on the anatomy of the subject in which the systems and devices will be used, the size and shape of components with which the systems and devices will be used, and the methods and procedures in which the systems and devices will be used.

(11) The systems, devices, and methods disclosed herein can be implemented using a robotic surgical system. WIPO Patent Publication No. WO 2014/151621 filed on Mar. 13, 2014 and entitled “Hyperdexterous Surgical System” is incorporated by reference.

(12) In general, a surgical system is described that allows a user (e.g., a surgeon) to control the operation of surgical instruments. In particular, the surgical instruments controlled by the described system are advanced surgical tools, for example, endocutters, and/or tools that have functionalities including at least one of cutting tissue, stapling tissue, and delivering energy to tissue. The controls of advanced tools may not be easily mapped to those of standard robotic tools (e.g., graspers, needle drivers etc.). Therefore, there is a need to develop control systems that allow a user to seamlessly control advanced surgical tools during surgery. The systems and methods described herein, however, are not limited to advanced surgical tools, and can be applied to virtually any surgical instrument used in a surgical procedure, including robotic tools.

(13) During a minimally invasive procedure, or any surgical procedure in which the surgical instrument is outside of the surgeon's natural field of view, an image of the surgical environment as well as relevant portions of the surgical instrument is typically generated and displayed on a display to the surgeon, such as on a video monitor. Such an image is typically displayed in real-time. In aspects of the system described herein, the image of the surgical instrument is modified to include a virtual control panel. The surgeon (or any other user) can interact with the control panel through a user interface device (e.g., one that is able to move in 3D space). The control panel can be pinned to a fixed location in the display, or can be moved to various locations in the modified image. For example, the location of the control panel can be fixed relative to a portion of the surgical instrument (e.g., shaft of a surgical instrument), and the control panel can follow the surgical instrument as it moves in the modified image. The control panel can change in size, shape, aspect ratio, and perspective to match the 2D or 3D orientation of the surgical instrument and/or the surrounding environment. Moreover, the control panel can appear in two dimensions (if viewed on a 2D display) or in three dimensions ((if viewed stereoscopically via headset, goggles, or glasses). The control panel can be configured to eliminate or minimize any obstruction to the field of view

of the surgical environment. For example, portions of the control panel can be transparent or semi-transparent, and the degree of transparency can be adjustable. Alternately or additionally, the surgeon can change the location of the control panel in the modified image, for example, by dragging and dropping the control panel to a desired location.

(14) A control panel can be customized for the surgical system and/or the instrument and/or for a particular procedure or use of the surgical system or instrument. For example, the operating parameters of an endocutter are different from those of a needle driver or an energy-delivering tool, and therefore, the control panel of an endocutter may be different from that of needle driver or an energy-delivering tool. Additionally, the relevant operating parameters of a surgical instrument may change based on the needs of a surgical procedure at any given time. For example, once an endocutter has been positioned in a desired location and clamped on a target tissue, the relevant operating parameters may include articulation angle, knife position, timing of jaw opening, and firing etc. The image of the control panel that is displayed after the endocutter has been clamped onto tissue will include the control for the operating parameters mentioned above.

(15) FIG. 1 is a perspective view of one embodiment of a surgical robotic system **100** that can be used in telesurgery. The system **100** includes a patient-side portion **110** that is positioned adjacent to a patient **112**, and a user-side portion **111** that is located a distance from the patient, either in the same room and/or in a remote location. The patient-side portion **110** generally includes one or more robotic arms **120** and one or more tool assemblies **130** that are configured to releasably couple to a robotic arm **120**. The user-side portion **111** generally includes a vision system **113** for viewing the patient **112** and/or surgical environment, and a control system **115** for controlling the movement of the robotic arms **120** and each tool assembly **130** during a surgical procedure.

(16) The control system **115** can have a variety of configurations and it can be located adjacent to the patient, e.g., in the operating room, remote from the patient, e.g., in a separate control room, or it can be distributed at two or more locations. As an example of a dedicated system, a dedicated system control console can be located in the operating room, and a separate console can be located at a remote location. The control system **115** can include components that enable a user to view a surgical environment of a patient **112** being operated on by the patient-side portion **110** and/or to control one or more parts of the patient-side portion **110** (e.g., to perform a surgical procedure at the surgical environment **112**). In some embodiments, the control system **115** can also include one or more manually-operated input devices, such as a joystick, exoskeletal glove, a powered and gravity-compensated manipulator, or the like. These input devices can control teleoperated motors which, in turn, control the movement of the surgical system, including the robotic arms **120** and tool assemblies **130**.

(17) The patient-side portion can also have a variety of configurations. As depicted in FIG. 1, the patient-side portion **110** can couple to an operating table **114**. However, in other embodiments, the patient-side portion **110** can be mounted to a wall, to the ceiling, to the floor, or to other operating room equipment. Further, while the patient-side portion **110** is shown as including two robotic arms **120**, more or fewer robotic arms **120** can be included. Furthermore, the patient-side portion **110** can include separate robotic arms **120** mounted in various positions, such as relative to the surgical table **114**. Alternatively, the patient-side portion **110** can include a single assembly that includes one or more robotic arms **120** extending therefrom.

(18) FIG. 2 is a schematic view of an example of surgical system **200** configured to generate modified images of a surgical environment (e.g., surgical instrument **204**, target tissues, tissues surrounding target tissues, etc.) in real-time. The surgical system **200** includes a surgical instrument **204**, a controller **202** to control the operation of the surgical instrument **204**, a camera module **210** configured to capture visible-light images of the surgical instrument **204**, and relay one or more signals related to the captured image to a processor **216**. The processor **216** can also communicate with the controller **202**. For example, the processor can receive operating parameters of the surgical instrument **204** from the controller **202**, and transmit to the controller **202** control signals that can

change the operating parameters. The processor **216** can communicate with a database **218** which can store information related to the surgical instrument **204** (e.g., control panel information). The processor **216** can generate a modified image that includes the visible-light image from the camera **210**, and a virtual control panel that allows a user to control the operating parameters of the surgical instrument **204**. The modified image can be displayed on a display **222**, which can be wall or table mounted or on an accessory worn by the surgeon. The surgical system **200** also includes an input device **224** which can communicate with the processor **216**. A user (e.g., surgeon) can interact with the modified image (e.g., change operating parameters in the control panel) using the input device **224**. Signals in the surgical system **200** (e.g., between camera module **210** and processor **216**, controller **202** and processor **216**, input device **224** and processor **216**, etc.) can be communicated wirelessly (Bluetooth, WiFi, etc.) or through a data cable (e.g., optical fiber, coaxial cable, etc.).

(19) A light source (not shown) can generate visible light which is reflected by the surgical environment. The reflected light can be visible light (e.g., having a wavelength of about 400 nm to 800 nm) or light of a wavelength that is outside of the visible spectrum (e.g., infrared and ultraviolet light). A portion of the reflected visible-light (e.g., having a wavelength of about 400 nm to 800 nm) is captured by the camera module **210**. The camera module **210** comprises a lens **206** configured to focus visible-light onto a detector **212**. The quality of the visible-light image can be improved, for example, by placing detector **212** in the focal plane of the lens **206**.

(20) The detector **212** is able to detect light reflected by the surgical instrument. As shown in FIG. 3, an exemplary detector **212** comprises an array of photosensitive sites (e.g., **212a-c**, etc.), which can absorb electromagnetic radiation impinging on the site, and generate an electrical signal (e.g., voltage signal, current signal, etc.) that is representative of the impinged radiation. For example, the strength of the electrical signal can be proportional to the intensity of the impinged electromagnetic radiation. Photosensitive sites typically have a spectral range which determines the range of frequencies that can be efficiently detected by the site. For example, a silicon (Si) photosensitive site can detect visible to near infrared radiation (spectral range 400-1000 nm), and a germanium (Ge) or indium gallium arsenide (InGaAs) photosensitive site can detect near infrared radiation (spectral range 800-2600 nm). A suitable type of photosensitive site that is appropriate for the spectral range of the electromagnetic radiation that one wants to detect can be selected by a person skilled in the art.

(21) A photosensitive site can be configured to detect a desired wavelength (or a narrow range of wavelength around the desired wavelength) of electromagnetic radiation that lies within its spectral range by using an optical filter. The optical filter, which is placed in the path of the electromagnetic radiation directed towards the photosensitive site, filters out all radiation except for that corresponding to the desired wavelength. For example, a Si photosensitive site (e.g., **212a**) with a green color filter will primarily detect green light (approximately 500 nm).

(22) In one example a detector (e.g., detectors **212**) detects an image of the surgical environment by combining the images of different regions of the object captured by various photosensitive sites in the detector. When the light reflected by the surgical instrument impinges on the detector **212**, a photosensitive site therein (e.g., **212a**, **212b**, **212c**, etc.) detects a part of the reflected light that represents an image of a region of the surgical instrument. The photosensitive site then generates an electrical signal that is representative of the captured image. This electrical signal is converted to a digital signal by an analog-to-digital converter (ADC). The digital signal has discretized values that represent, for example, the intensity of the detected radiation. As will be described below, the digital signal can also include information related to the frequency (color) of the detected radiation. The values of the digital signals from the various photosensitive sites (collectively referred to as an image dataset) are representative of the image of the surgical instrument. There can be a one-to-one relationship between a digital signal value stored in the image dataset, and the photosensitive site that has produced the digital signal value (e.g., the digital signal value can include information that identifies photosensitive site that has generated the digital signal). Therefore, by identifying a

digital signal value in the image dataset the photosensitive site that generated the digital value can be identified (or vice-versa). The processor then generates the image of the surgical environment from the image dataset that can be displayed on a display device **222** (e.g., a monitor). Each pixel in the display device can represent one digital signal value in the image dataset. In other words, each pixel in the display device can represent the radiation detected by a unique photosensitive site in the detector **212**.

(23) A colored image of a surgical environment can be generated by placing optical filters (or an array of optical filters) in the path of the electromagnetic radiation directed towards a detector. For example, an array of color filters (e.g., Bayer filter, RGBE filter, CYYM filter, CYGM filter, etc.) can be placed before an array of photosensitive sites. As a result, each photosensitive site receives electromagnetic radiation of a particular wavelength (or color). For example, for a Bayer filter, each photosensitive site detects one of red, blue or green color. The processor can use a demosaicing algorithm to process an image dataset obtained using a Bayer filter to generate a “full-color” image (i.e., an image with multiple colors).

(24) If a light optical filter is placed before the first detector **212**, it will detect a light image of the surgical environment. As a result, a light image dataset is generated (as described above) and transmitted to the processor **216**. The light image dataset can include information related to the intensity and wavelength (color) of the detected light for each photosensitive site.

(25) The controller **202** is configured to control the operation of the surgical instrument **204**. For example, if the surgical instrument **204** is an endocutter, the controller can change its operating parameters (e.g., articulation angle, knife position, timing of opening and firing, etc.). The controller **202** can also control the energy delivered by the surgical instrument **204** to a target tissue. For example, the controller **202** may drive an ultrasonic transducer in the surgical instrument and/or generate radio frequency energy which the surgical instrument can deliver to a tissue through an energy-delivering end effector. The controller **202** is configured to receive control signals from the processor **216** and such control signals can change the operating parameters of the surgical instrument **204**. The controller **202** can transmit a signal to the processor **216** that includes information related to the operating parameters and identity of the surgical instrument **204**. The signal may be transmitted when the controller **202** receives a request signal from the processor **216** requesting information related to the surgical instrument **204**. The controller **202** can include a memory device that has a log file which keeps track of the operating parameters of the surgical instrument **202**. The log file (or a part thereof) can be transmitted to the processor **216**. In some embodiments, the processor **216** and the controller **202** may not communicate directly. For example, their communication may be routed through one or more devices (e.g., processors, routers etc.).

(26) The processor **216** can identify the surgical instrument **204** and its operating parameters, for example, based on the signal from the controller **202**. Processor **216** can access information associated with the surgical instrument **204** that may be stored in the database **218** or that may be communicated to the processor from the surgical instrument itself. For example, the processor may search the database **218** for a template of a virtual control panel related to surgical instrument **204**, or it can receive such a template from the surgical instrument. The database may also contain a control panel template that has been customized by the user based on their specific requirements. The choice of the template may also be based on the operating parameters of the surgical instrument **204**. The processor **216** can modify the light image of the surgical environment obtained from the camera module **210** by superimposing the control panel **232** onto the image. The modified image can be displayed on the display **222**.

(27) As shown in FIG. 2, the modified image includes the image of the surgical environment **230** and the control panel **232**. As explained above, the control panel can be located at a fixed location on the display **222** (e.g., top left, top right, bottom left, bottom right, etc.) or the location of the control panel can change, such as based on user commands. Moreover, a user can interact with the

control panel **232** through the input device **224**. For example, if the input device **224** is a user interface device, processor **216** can generate a pointer in the modified image that can track the motion of the user interface device. The user can use the user interface device to scroll, drag, actuate (e.g., click, double click, etc.) on various control elements in the control panel. If the user is operating in a 3D environment (i.e., via a stereoscopic camera system) the input device **224** may appear to the user as a 3D pointer that moves within the 3D space surrounding the device and/or control panel. Based on this input, the processor **216** can generate a control signal to change one or more operating parameters, and such a control signal is transmitted to the controller **202**.

(28) FIG. **4** illustrates an example of the control panel **232** when the surgical instrument **204** is an endocutter. As shown, the control panel **232** comprises various control elements or buttons (**402-414**). The user may open or close the jaws of the endocutter by actuating virtual buttons **402**, **404**, respectively. The user may change the articulation angle to the left or right by actuating virtual buttons **410a** and **410b**, respectively, and button **408** can be actuated to return the tool to its home position. Window **412** within control panel **232** indicates the degree of articulation associated with the surgical instrument. In another functionality, the user can control the knife action by actuating fire button **406** to cause the knife to fire. Part of the virtual control panel also includes a knife position tab **414**. In one embodiment the control elements in windows **412**, **414** can be used to change the articulation angle and the knife position, respectively, although it is understood that windows **412**, **414** can be configured only to display information and not to enable control of the instrument.

(29) As noted above, a user can vary the degree of transparency of the control panel **232**. For example, if the transparency of the control panel is lowered, the visibility of the surgical environment behind the control panel decreases and vice versa. The user may want to adjust the transparency of the control panel, for example, when the control panel is overlaid on an area of the surgical environment that needs to be monitored (e.g., target tissue). Alternately, the user may move the control panel in the modified image (e.g., by dragging and dropping using the input device **224**).

(30) As described above, in some embodiments of the surgical system **200**, the control panel **232** can follow the motion of the surgical instrument **204** in the modified image. The processor **216** can be configured to identify the image of the surgical instrument **204** from the image of the surgical environment (obtained from the camera module **210**), and superimpose the control panel **232** at a fixed position relative to the image of the surgical instrument. As the surgical instrument **204** moves, the control panel **232** follows its motion. FIG. **5** illustrates an example of a modified image **500** of an endocutter **502** performing a surgical procedure on a target tissue **506**. The control panel **504** is located near the distal end of the endocutter so that it is most easily viewed by a surgeon. The control panel **506** can be configured to appear after the endocutter is locked to the location, for example, by the release of a clutch pedal by the user. In another embodiment, the control panel **232** can be located on or near the shaft of the endocutter.

(31) The processor can identify data related to the image of the surgical instrument **204** (e.g., endocutter) from the reflected light image dataset generated by the camera module **210**. In one embodiment, the data is identified based on a predetermined relative position between the first detector **212** and the surgical instrument **204**. In this embodiment, the camera module **210** is attached to the surgical instrument **204** such that the relative position of the surgical instrument with respect to the camera module **210** remains fixed. This can be done, for example, by having a mounting feature on the surgical instrument to which the camera module **210** can be removably attached. Devices (detector **212**, lens **206**, etc.) inside the camera module **210** can be positioned in a predetermined configuration. Alternately, the devices can be attached to piezoelectric actuators that allow them to move relative to one another. This can allow the detector to detect a sharp image of the surgical instrument. For example, it can be desirable to place the detector **212** in the focal plane of the lens **206**. Mechanical movements and thermal expansion of the camera module **210** and the

devices therein can move the detectors out of the focal plane of the lens. The detectors can be adjusted back into the focal plane by the piezoelectric actuators that can be controlled by the processor **216**, or by an input from a user. The surgical instrument **204** and the camera module **210** (and the devices inside the camera module) can be adjusted to a desired predetermined position prior to the insertion of camera module **210** and surgical instrument **204** in the surgical environment. The photosensitive sites in detectors **212** that capture the image of the surgical instrument **204** can be identified based on the predetermined orientation of the detector **212** and the surgical instrument **204**. Information related to the location of the aforementioned photosensitive sites can be stored in the database **218**. The processor **216** can identify surgical instrument image data in the reflected light image dataset. This can be done, for example, by arranging the image data, captured by the photosensitive sites, in a predetermined pattern in the light image. For example, the image data captured by the photosensitive site **212a** can be placed at a predetermined location in the visible-light dataset. Information about this relationship can be stored in an index data file in the database **218**. Based on the index data file, the processor **218** can identify the image data (from the reflected image dataset) corresponding to the image detected by the photosensitive site **212a**. Alternately, the image data can include information that identifies the photosensitive site that generated it.

(32) In another embodiment, the surgical instrument is identified in the light image based on one or more markers on the surgical instrument **204**, and multiple cameras **610**, **612**, **614**, **616** are used to image the surgical environment. As shown in FIG. **6**, the surgical instrument **204** includes a number of markers **602**, **604**, **606**, **608** located on its surface. In one embodiment the markers are regions on the surgical instrument **204** that reflect electromagnetic radiation of a given frequency. For example, the markers can be configured to reflect light of a certain color. The color of the markers can be selected such that the color is not naturally present in the surgical environment (e.g., green, blue, etc.). The processor **216** thus identifies photosensitive sites that detect the image of the markers based on marker color. As described above, the reflected light image dataset can include, for each photosensitive site, information related to the color of the detected light. The processor can search in the reflected light image dataset for data representative of the color of the marker. The processor **216** identifies the markers **602**, **604**, **606**, **608** (and therefore the relative positions of the markers) in the reflected light image and compares this information with data from a database of surgical instruments stored in the database **218**. The database includes information related to marker color and marker position for various surgical instruments. Additionally, for a given surgical instrument, the database may include information related to the relative position of the markers in an image of the surgical instruments from multiple viewpoints. For example, the relative positions of the markers **602**, **604**, **606**, **608** in the image of the surgical instrument **204** in the embodiment of FIG. **6** will depend on the relative position of the camera (e.g., **610**, **612**, **614**, **616**) that captures the image.

(33) The processor can use an image recognition algorithm to identify the data in the visible-light dataset that represents the image of the surgical instrument. In one example the image recognition algorithm receives input information related to the position of the markers in the captured image, and information related to various surgical instruments stored in the database **218**. The processor compares the relative positions of markers in the captured image with the orientation information of markers of the devices in the database **218** by using various pattern recognition techniques. Based on this comparison, the processor can identify data representative of image of the surgical instrument in the visible-image dataset. Additionally or alternatively, the processor can use a pattern recognition algorithm and/or device database to work with devices that are not readily known to or controlled by the robotic surgical system (i.e., a hand-operated stapler, retractors, etc.) It is to be understood that markers may comprise a specific geometric shape and/or color, either as something that is applied to the device (e.g., green dots) or the color can be inherent of typical devices (e.g., silver straight jaws and blackened shaft).

(34) After the processor **216** identifies the image of the surgical instrument **204**, it can provide the user of the surgical system **200** with an option to choose the location of the control panel **232**. The user may drag the control panel **232** to a desired orientation relative to the image of the surgical instrument in the modified image (e.g., at the proximal or distal ends of the surgical instrument, shaft of the surgical instrument, etc.) and lock the position of the control panel **232**. As the image of the surgical instrument **204** moves in real-time, the orientation of the control panel **232** is accordingly adjusted to maintain the locked relative orientation.

(35) One or more aspects or features of the subject matter described herein can be realized in digital electronic circuitry, integrated circuitry, specially designed application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs) computer hardware, firmware, software, and/or combinations thereof. These various aspects or features can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which can be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device. The programmable system or computer system may include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

(36) The computer programs, which can also be referred to as programs, software, software applications, applications, components, or code, include machine instructions for a programmable processor, and can be implemented in a high-level procedural language, an object-oriented programming language, a functional programming language, a logical programming language, and/or in assembly/machine language. As used herein, the term “machine-readable medium” refers to any computer program product, apparatus and/or device, such as for example magnetic discs, optical disks, memory, and Programmable Logic Devices (PLDs), used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. The machine-readable medium can store such machine instructions non-transitorily, such as for example as would a non-transient solid-state memory or a magnetic hard drive or any equivalent storage medium. The machine-readable medium can alternatively or additionally store such machine instructions in a transient manner, such as for example as would a processor cache or other random access memory associated with one or more physical processor cores.

(37) To provide for interaction with a user, one or more aspects or features of the subject matter described herein can be implemented on a computer having a display device, such as for example a cathode ray tube (CRT) or a liquid crystal display (LCD) or a light emitting diode (LED) monitor for displaying information to a user and a keyboard and a pointing device, e.g., a mouse, a trackball, etc., by which a user may provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well. For example, feedback provided to a user can be any form of sensory feedback, such as for example visual feedback, auditory feedback, or tactile feedback; and input from a user may be received in any form, including, but not limited to, acoustic, speech, or tactile input. Other possible input devices include, but are not limited to, touch screens or other touch-sensitive devices such as single or multi-point resistive or capacitive trackpads, voice recognition hardware and software, optical scanners, optical pointers, digital image capture devices and associated interpretation software, and the like.

(38) FIG. 7 illustrates an exemplary embodiment of a computer system **700**. As shown, the computer system **700** includes one or more processors **702** which can control the operation of the computer system **700**. “Processors” are also referred to herein as “controllers.” The processor(s) **702** can include any type of microprocessor or central processing unit (CPU), including

programmable general-purpose or special-purpose microprocessors and/or any one of a variety of proprietary or commercially available single or multi-processor systems. The computer system **700** can also include one or more memories **704**, which can provide temporary storage for code to be executed by the processor(s) **702** or for data acquired from one or more users, storage devices, and/or databases. The memory **704** can include read-only memory (ROM), flash memory, one or more varieties of random access memory (RAM) (e.g., static RAM (SRAM), dynamic RAM (DRAM), or synchronous DRAM (SDRAM)), and/or a combination of memory technologies. (39) The various elements of the computer system **700** can be coupled to a bus system **712**. The illustrated bus system **712** is an abstraction that represents any one or more separate physical busses, communication lines/interfaces, and/or multi-drop or point-to-point connections, connected by appropriate bridges, adapters, and/or controllers. The computer system **700** can also include one or more network interface(s) **706**, one or more input/output (IO) interface(s) **708**, and one or more storage device(s) **710**.

(40) The network interface(s) **706** can enable the computer system **700** to communicate with remote devices, e.g., other computer systems, over a network, and can be, for non-limiting example, remote desktop connection interfaces, Ethernet adapters, and/or other local area network (LAN) adapters. The IO interface(s) **708** can include one or more interface components to connect the computer system **700** with other electronic equipment. For non-limiting example, the IO interface(s) **708** can include high speed data ports, such as universal serial bus (USB) ports, 1394 ports, Wi-Fi, Bluetooth, etc. Additionally, the computer system **700** can be accessible to a human user, and thus the IO interface(s) **708** can include displays, speakers, keyboards, pointing devices, and/or various other video, audio, or alphanumeric interfaces. The storage device(s) **710** can include any conventional medium for storing data in a non-volatile and/or non-transient manner. The storage device(s) **710** can thus hold data and/or instructions in a persistent state, i.e., the value(s) are retained despite interruption of power to the computer system **700**. The storage device(s) **710** can include one or more hard disk drives, flash drives, USB drives, optical drives, various media cards, diskettes, compact discs, and/or any combination thereof and can be directly connected to the computer system **700** or remotely connected thereto, such as over a network. In an exemplary embodiment, the storage device(s) can include a tangible or non-transitory computer readable medium configured to store data, e.g., a hard disk drive, a flash drive, a USB drive, an optical drive, a media card, a diskette, a compact disc, etc.

(41) The elements illustrated in FIG. 7 can be some or all of the elements of a single physical machine. In addition, not all of the illustrated elements need to be located on or in the same physical machine. Exemplary computer systems include conventional desktop computers, workstations, minicomputers, laptop computers, tablet computers, personal digital assistants (PDAs), mobile phones, and the like.

(42) The computer system **700** can include a web browser for retrieving web pages or other markup language streams, presenting those pages and/or streams (visually, aurally, or otherwise), executing scripts, controls and other code on those pages/streams, accepting user input with respect to those pages/streams (e.g., for purposes of completing input fields), issuing HyperText Transfer Protocol (HTTP) requests with respect to those pages/streams or otherwise (e.g., for submitting to a server information from the completed input fields), and so forth. The web pages or other markup language can be in HyperText Markup Language (HTML) or other conventional forms, including embedded Extensible Markup Language (XML), scripts, controls, and so forth. The computer system **700** can also include a web server for generating and/or delivering the web pages to client computer systems.

(43) In an exemplary embodiment, the computer system **700** can be provided as a single unit, e.g., as a single server, as a single tower, contained within a single housing, etc. The single unit can be modular such that various aspects thereof can be swapped in and out as needed for, e.g., upgrade, replacement, maintenance, etc., without interrupting functionality of any other aspects of the

system. The single unit can thus also be scalable with the ability to be added to as additional modules and/or additional functionality of existing modules are desired and/or improved upon.

(44) A computer system can also include any of a variety of other software and/or hardware components, including by way of non-limiting example, operating systems and database management systems. Although an exemplary computer system is depicted and described herein, it will be appreciated that this is for sake of generality and convenience. In other embodiments, the computer system may differ in architecture and operation from that shown and described here.

(45) Preferably, components of the invention described herein will be processed before use. First, a new or used instrument is obtained and if necessary cleaned. The instrument can then be sterilized. In one sterilization technique, the instrument is placed in a closed and sealed container, such as a plastic or TYVEK bag. The container and instrument are then placed in a field of radiation that can penetrate the container, such as gamma radiation, x-rays, or high energy electrons. The radiation kills bacteria on the instrument and in the container. The sterilized instrument can then be stored in the sterile container. The sealed container keeps the instrument sterile until it is opened in the medical facility.

(46) Typically, the device is sterilized. This can be done by any number of ways known to those skilled in the art including beta or gamma radiation, ethylene oxide, steam, and a liquid bath (e.g., cold soak). An exemplary embodiment of sterilizing a device including internal circuitry is described in more detail in U.S. Pat. No. 8,114,345 filed Feb. 8, 2008 and entitled "System And Method Of Sterilizing An Implantable Medical Device." It is preferred that device, if implanted, is hermetically sealed. This can be done by any number of ways known to those skilled in the art.

(47) One skilled in the art will appreciate further features and advantages of the invention based on the above-described embodiments. Accordingly, the invention is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated herein by reference in their entirety.

Claims

1. A robotic surgical system, comprising: a tool assembly configured to be removably coupled to a robotic arm and comprising a shaft extending distally from a housing and an end effector coupled to the shaft, the end effector being configured to treat a target tissue; a camera system configured to detect light reflected by at least one of the end effector and the shaft and generate a first signal comprising a first dataset representative of an image of the at least one of the end effector and the shaft; and a processor configured to: receive the first signal, generate, after the receipt of the first signal, a modified image of the at least one of the end effector and the shaft that includes a control panel, wherein the control panel comprises one or more control elements representative of one or more operating parameters of the robotic surgical system and one or more control buttons configured to control one or more operations of the tool assembly when actuated by a user, and receive an input to the control panel from the user, the input being effective to fix a location of the control panel on the modified image relative to a feature of the tool assembly such that the control panel is configured to follow the feature of the tool assembly in the modified image as the tool assembly moves.
2. The robotic surgical system of claim 1, wherein the processor is further configured to change at least one of a size, a shape, an aspect ratio, or a perspective of the control panel based on an orientation of the tool assembly.
3. The robotic surgical system of claim 1, wherein the processor is configured to adjust an orientation of the control panel as the modified image of the tool assembly moves in real-time.
4. The robotic surgical system of claim 1, wherein the feature is a distal portion of the shaft.
5. The robotic surgical system of claim 1, wherein the feature is an end effector of the tool assembly.

6. The robotic surgical system of claim 1, wherein the processor is further configured to avoid obstructing a field of view of a surgical environment with the control panel.
 7. The robotic surgical system of claim 6, wherein the processor is further configured to set at least a portion of the control panel as transparent or semi-transparent.
 8. The robotic surgical system of claim 1, wherein the modified image and the control panel are three-dimensional.
 9. The robotic surgical system of claim 1, wherein the control panel is configured to appear on the modified image after the tool assembly reaches the target tissue.
 10. A method, comprising receiving a first signal comprising a first dataset representative of an image of at least one of an end effector and a shaft of a tool assembly, wherein the end effector is configured to treat a target tissue; generating a modified image of the at least one of the end effector and the shaft that includes a control panel, wherein the control panel comprises one or more control elements representative of one or more operating parameters of a robotic surgical system and one or more control buttons configured to control one or more operations of the tool assembly when actuated by a user; and receiving an input to the control panel from the user, the input being effective to set a location of the control panel relative to a feature of the tool assembly such that the control panel follows the feature of the tool assembly in the modified image as the tool assembly moves.
 11. The method of claim 10, further comprising changing at least one of a size, a shape, an aspect ratio, or a perspective of the control panel based on an orientation of the tool assembly.
 12. The method of claim 10, further comprising adjusting an orientation of the control panel as the modified image of the tool assembly moves in real-time.
 13. The method of claim 10, wherein the feature is a distal portion of the shaft.
 14. The method of claim 10, wherein the feature is an end effector of the tool assembly.
 15. The method of claim 10, further comprising avoiding obstructing a field of view of a surgical environment with the control panel.
 16. The method of claim 15, further comprising setting at least a portion of the control panel as transparent or semi-transparent.
 17. The method of claim 10, wherein the modified image and the control panel are three-dimensional.
 18. The method of claim 10, wherein the control panel is configured to appear on the modified image after the tool assembly reaches the target tissue.
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